

Quantitative easing: a sceptical survey

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Abstract Evaluation of quantitative easing (QE) is difficult as it is only used in response to severe and unusual economic difficulties. Despite this, we argue that two main conclusions can be drawn from a sceptical reading of the evidence. First, large-scale asset purchases reduce government bond rates, especially at the longer end of the yield curve. However, this effect may be temporary and is small if bond rates are already low, while initial waves of QE are more effective than subsequent programmes. Second, QE appears to have been effective in late 2008 and 2009, preventing even larger declines in output and inflation than were experienced. We argue that the literature is limited, relying on similar methodologies and largely originating in central banks. Exploration of alternative approaches to QE would be useful in widening an evidence base that is currently too narrow.

Key words: quantitative easing (QE), financial crisis

JEL classification: E43, E58

I. Introduction

Assessment of the impact of quantitative easing (QE) is a difficult task. QE programmes are both very unusual and very large. They are only undertaken in times of severe economic difficulty and in circumstances when more conventional monetary policy cannot be used because the policy rate is at the zero lower bound. This severely limits investigation of the effects of QE. Studies can use contemporary high-frequency data from periods when QE is in operation, in which case data on some key macroeconomic variables, most especially output and inflation, are not available; or else they can use data from periods when QE is not in operation, in which case they must assume that key macroeconomic relationships are not disrupted by the extraordinary events that led QE to be adopted. Clearly, neither approach is ideal.

QE consists of large purchases of mainly longer-term government bonds and related assets. Policy-makers in the US and UK turned to QE in the hope that these purchases

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would reduce interest rates faced by households and firms and result in increased output and inflation. The structure of this survey reflects this chain of causation. We first review evidence on the impact of QE on government bond rates. We consider evidence from the 'QE1' and 'QE2' programmes as well as 'Operation Twist' in the US and evidence from the UK 'QE1', 'QE2', and 'QE3' programmes. In doing so, we review the results of event studies and estimates of econometric models. We then review evidence on the impact of QE on other asset prices, such as corporate bond rates, equity yields, exchange rates, and commodity prices. Finally, we consider evidence on the impact of QE on the 'real economy', reviewing econometric studies of the impact of QE on output and inflation.

We review empirical evidence on the impact of QE programmes in the US and UK since 2008. We do not consider the impact of earlier episodes of QE, especially in Japan, and do not consider interventions by the European Central Bank (ECB) since 2007, as these focused on supporting short-term money markets and so are different from the large-scale asset purchases seen in the US and UK. We do not consider alternative policies to address the economic crisis or political economy aspects of QE. We do not address the causes of the financial crisis or the possible future shape of policy including macroprudential regulation.

II. Evidence of the impact of QE on government bond rates

(i) Event studies

Event studies assume that all changes in the variable of interest that occur within a short time period are due to the event under consideration. Most evidence comes from the US QE1 programme. The first stage of the programme, \$600 billion purchases of housing agency debt and related securities, was announced on 25 November 2008 (although arguably the programme was signalled in Chairman Bernanke's Jackson's Hole speech in late August). Following several indications of a likely increase in the size of the programme, a major expansion of \$1.15 trillion, comprising \$850 billion additional purchases of agency debt and \$300 billion longer-dated government bonds, was announced on 18 March 2009. Movements in the government bond rate in the days after these announcements have been taken as evidence of the impact of the programme. Three studies find that the programme reduced 10-year government bond yields by around 100 basis points (bp), with the 18 March 2009 announcement having a particularly large effect. [Gagnon *et al.* \(2011\)](#) report a reduction of 91 bp, [Glick and Leduc \(2011\)](#) find an effect of around 100 bp, and [Krishnamurthy and Vissing-Jorgensen \(2011\)](#) report a reduction of 107 bp. Another study, [Meaning and Zhu \(2011\)](#), reports a smaller but still sizeable effect of around 80 bp; they also report reductions of around 85 bp in the 5-year bond rate.

The first round of QE in the UK, UK QE1, began gradually. The body used for asset purchases, the Asset Purchase Facility Fund, was set up in January 2009 and some purchases of commercial bills occurred in February. The Inflation Report news conference on Wednesday 11 February 2009 indicated that an asset purchase programme

was likely. On 5 March the Monetary Policy Committee (MPC) announced the purchase of £75 billion assets; subsequent announcements on 7 May, 6 August, and 5 November then increased the programme to £125 billion, £175 billion, and then £200 billion. Maintenance of the programme at £200 billion was announced in February 2010. Of the £200 billion purchases, £198 billion were of government bonds, spread across a wide range of maturities. Unlike US QE1, differing estimates of the impact of the programme have been reported. [Joyce *et al.* \(2011\)](#) find an overall reduction of around 100 bp in the 10-year government bond rate as a result of the programme, with the largest effect coming after the announcement on 5 March. [Meier \(2009\)](#) finds an effect of between 40 and 100 bp, arguing there is an ‘effect of QE on gilt yields of, at the very least, 35–60 basis points’. By contrast, [Glick and Leduc \(2011\)](#) and [Meaning and Zhu \(2011\)](#) find smaller effects, close to 50 bp. This may reflect the fact that these latter papers use a 1-day window whereas [Joyce *et al.* \(2011\)](#) use a 2-day window (and report a stronger response on the second day—[Joyce *et al.* \(2011, Chart 10\)](#)). The 50 bp difference between these studies highlights the importance of window length in event studies: adding a single day doubles the estimated impact of UK QE1.

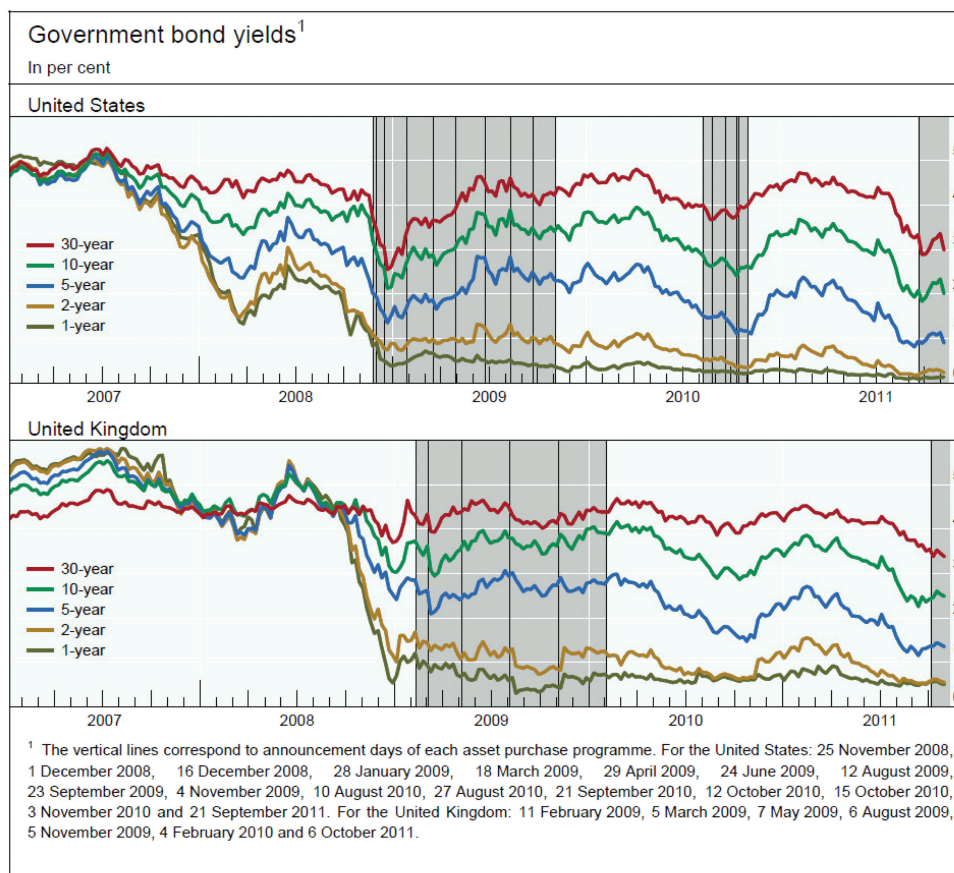
The first indication of the US QE2 programme was given on 10 August 2010 when the Federal Open Market Committee (FOMC) announced it would roll over holdings of Treasury securities as they matured, extending the life of the programme by using redemption payments to purchase fresh assets. Further statements led to the announcement on 3 November 2010 of \$600 billion purchases of long-dated government bonds. Evidence from event studies is mixed, but with no indication of a large impact. [Krishnamurthy and Vissing-Jorgensen \(2011\)](#) report a reduction of 18 bp in the 10-year government bond rate for a 1-day window and 30 bp for a 2-day window, with smaller reductions in the 5- and 30-year rates. [Glick and Leduc \(2011\)](#) find no effect on the 10-year rate, while [Meaning and Zhu \(2011\)](#) report a slight increase in the 10-year and a slight fall in the 5-year rate. The FOMC announced Operation Twist on 21 September 2011. This was an attempt to flatten the yield curve by selling \$400 billion assets with a maturity of less than 3 years and using the proceeds to purchase assets with a maturity over 6 years. The only event study on this, [Meaning and Zhu \(2011\)](#), finds this had a measurable but small impact, raising the 5-year rate but reducing the 10-year rate. This is consistent with evidence from a similar exercise in the early 1960s ([Swanson, 2011](#)).

The UK QE2 programme of £75 billion purchases of government bonds was announced on 6 October 2011. [Meaning and Zhu \(2011\)](#) report this did not reduce government bond rates, being associated with small increases in rates across the yield curve. The UK QE3 programme of a further £50 billion purchases was announced on 9 February 2012. We do a brief event study of the effects of this announcement. On 9 February, 20-, 10-, and 5-year (zero-coupon) UK government bond rates were 3.43 per cent, 2.40 per cent, and 1.11 per cent, respectively. On Friday 10 February, the day after the announcement, they were 3.36 per cent, 2.45 per cent, and 1.01 per cent, respectively. On Monday 13 February they were 3.37 per cent, 2.29 per cent, and 0.99 per cent, respectively. An event study with a 1-day window implies UK QE3 reduced the 20-year rate by 7 bp, increased the 10-year rate by 5 bp, and reduced the 5-year rate by 10 bp. A 2-day window implies UK QE3 reduced government bond rates, by 6 bp, 11 bp, and 12 bp respectively for 20-, 10-, and 5-year rates. These are small effects. The reductions

increased slightly in the rest of February 2012, but this may well reflect the negotiations around the Greek bail-out and bond-swap.

These event studies suggest that initial large-scale asset purchases were effective in reducing government bond rates in the US and UK. Given the scale of these programmes (by the end of UK QE1 the Bank of England held nearly 30 per cent of freely floating gilts) this is not surprising. Perhaps more surprising is the lack of impact from later initiatives: we might have expected more from the \$600 billion spent on the US QE2 programme. But event studies are only informative about movements in bond rates within a narrow time frame, no more than 6–10 days across a QE programme of many months. They do not address longer-term movements in bond rates. Figure 1 (reproduced from Meaning and Zhu (2011)) depicts UK and US government bond rates in 2007–11 with key QE-related events highlighted. US government bond rates fell over the period of the US QE1 programme, from close to 4 per cent in the 3rd quarter of 2008 to around 2 per cent in early 2009. Event studies suggest about half of this decline may have been due to QE1. However bond rates then rose sharply, ending 2009 close to their 2007 levels. This

Figure 1: Government bond yields in the US and the UK



Source: Meaning and Zhu (2011).

may mean that QE has only a temporary, and swiftly unwound, effect on bond rates. The fact that the most pronounced movements in US bond rates in this period occurred outside periods of QE, for example in the 3rd quarter of 2008, may indicate that other forces are more important. But this decline may also reflect anticipation of a QE programme; discussion of QE became widespread as the policy rate fell rapidly to its lower bound in the 3rd quarter; arguably the US QE1 programme was widely expected and the only 'news' in the initial announcement concerned the size of the programme.

The UK 10-year government bond rate also fell in the early months of the UK QE1 programme in February and March 2009, from 3.8 per cent to close to 3 per cent; anticipation of QE may have affected bond yields prior to this, especially after the launch of US QE 4 months earlier. The 100 bp reduction estimated in the event study in [Joyce *et al.* \(2011\)](#) suggests that the bond rate would have risen slightly in the absence of QE; the observed reduction is perhaps more consistent with the smaller 50 bp reduction estimated in [Meier \(2009\)](#), [Glick and Leduc \(2011\)](#), and [Meaning and Zhu \(2011\)](#), with the marked fall around 5 March especially suggestive of an effect. However, this reduction was soon reversed, and bond rates at the end of the programme were back to their levels of late 2008. As in the US, the strongest movements in UK government bond rates occurred outside periods of QE programmes. In particular, the periods leading up to bail-outs for Greece (May 2010 and July 2011; the latter renegotiated in October 2011 and February 2012), Ireland (November 2010), and Portugal (May 2011) are associated with falls in UK and US government bond rates, clearly suggestive of a 'flight-to-quality' around these events, irrespective of the effects of QE. Given that the scale of these bailouts was comparable to that of QE programmes (close to €240 billion for Greece, €85 billion for Ireland, and €78 billion for Portugal), this is not a surprise. This again raises the question of how much impact QE has on bond rates. Event studies cannot hope to address these issues, so we next consider econometric evidence.

(ii) Econometric studies

QE programmes are rare events that only happen in times of severe financial stress and economic turmoil. As a result, econometric investigation of QE requires difficult compromises that are reflected in the two types of econometric study of the impact of QE on bond yields. The first type uses data from periods when QE programmes were in operation. This enables direct investigation of these programmes but is only feasible using daily or other high-frequency data. The need for high-frequency data makes it difficult to control for variables such as inflation and output that might be expected to affect bond rates. This problem is reduced in normal circumstances when the short-term interest rate set by policy-makers reflects these factors. However, QE is not used in normal circumstances, only being used when the policy rate is at the zero lower bound and so unable to act as a high-frequency sufficient statistic for developments in the real economy. The inability to control for movements in inflation and output is a severe impediment for this type of study.

The second type of econometric study of QE uses lower frequency historical data from before 2008, using a richer set of explanatory variables. Typically, this approach studies the impact of changes in the stock of government bonds on bond rates in normal times. By scaling up responses to the relatively minor changes in the stock of bonds

in the pre-crisis period to reflect the much greater movements observed during QE, some indication of the likely impact of QE may be obtained. There are two apparent weaknesses with this approach. First, it can indicate the likely impact of a generic QE initiative, but not the impact of a specific QE programme. Second, and more seriously, it assumes that the relationships that drive bond rates in normal circumstances are not changed by the financial stress and economic turmoil that lead to QE being used or by the very large scale of QE programmes. These are both debatable assumptions.

As with event studies, most evidence is available for the US QE1 programme. Taken together, econometric estimates suggest the programme did reduce government bond rates, especially 10–15 year rates. However, the effects are smaller and more diverse than those found in event studies. [Gagnon *et al.* \(2011\)](#) use historical monthly data for 1985–2008; they regress the 10-year Treasury bond yield and the 10-year yield term premium on official holdings of US Treasury debt (the amount of Treasury debt held by the US Treasury, the Federal Reserve, and foreign governments and central banks) including the unemployment gap, core consumer price index (CPI) inflation, and interest-rate volatility as controls. They estimate that increased official holdings of Treasury debt reduce the term premium and hence drive down government bond rates. Using these estimates to infer the impact of the \$1.75 trillion US QE1 programme, they obtain a reduction in the 10-year bond rate of between 38 and 82 bp, depending on the dependent variable. [D'Amico and King \(2010\)](#) use the contemporary data approach, using daily data for 25 March to 30 October 2009 to estimate yield-curve models. They regress the change in an asset's yield since the start of the sample on US QE1 purchases of the asset and purchases of substitute assets, controlling for changes in the residual maturity of the asset. Doing this for a range of assets they find significant reductions in bond yields across the yield curve, with a largest effect of around 50 basis points on the 15-year bond rate, a slightly smaller effect on the 10-year rate, but reductions of less than 20 bp for durations of less than 5 years or over 25 years.

[Wright \(2011\)](#) constructs measures of the 'surprise' component of various US monetary policy announcements; he finds, for example, that the 25 November 2008 initial \$600 billion announcement of US QE1 is equivalent to a 80 bp cut in the policy rate, while the major extension of US QE1 to \$1.75trillion announced on 18 March 2009 is equivalent to a 340 bp reduction. [Glick and Leduc \(2011\)](#) evaluate the impact of QE programmes by regressing government bond yields on this measure using daily data. For US QE1, they estimate a significant reduction in the 10-year bond rate. However, since the estimates are expressed in terms of the effect of a 1-standard deviation monetary policy shock, it is unclear what this implies for the size of the reduction in bond rates due to US QE1. [Szczerbowicz \(2011\)](#) considers the different aspects of US QE1, allowing purchases of government bonds and mortgage-backed securities (MBS) and agency debt to have different effects on the 10-year rate. She finds similar effects from both components of QE1, a reduction of 22 bp from government bond purchases and 17 bp from purchases of MBS and agency debt. [Meaning and Zhu \(2011\)](#) use a similar approach to [D'Amico and King \(2010\)](#). For US QE1 they find an average reduction of 27 bp for bonds with maturities between 5 and 25 years. However, they only consider the effect of the \$300 billion purchase of government bonds; since these were less than 20 per cent of the amount purchased, this is not an estimate of the impact of the whole programme.

Considering the UK QE1 programme, [Joyce *et al.* \(2011\)](#) estimate VAR (vector autoregression) and multivariate GARCH (generalized autoregressive conditional

heteroskedasticity) models using monthly historical data for the 1991–2007 period. Using a portfolio balance approach and a VAR comprising returns on and portfolio shares of gilts, investment-grade corporate bonds, equities, and money as endogenous variables and inflation, industrial production, and the slope of the yield curve as exogenous variables (this is a strong assumption since it rules out any impact of the various endogenous interest rates on output and inflation), they simulate the effects of QE by considering a shock that reduces the share of government bonds in private-sector portfolios, calibrated to mimic the scale of UK QE1. They estimate this shock reduces yields on gilts by 85 bp. However, the impulse responses unwind rapidly and the effect after 6 months is reduced to 32 bp. This rapid reduction in the estimated impact of QE might suggest some caution in interpreting impact studies that focus on immediate responses to announcements. Using contemporary daily data and the same methodology as they used to assess US QE1, [Glick and Leduc \(2011\)](#) find that UK QE1 also led to a significant reduction in the 10-year bond rate, although the size of the effect is unclear. [Meaning and Zhu \(2011\)](#) use contemporary daily data to estimate a model that lacks the rich dynamic responses of *Joyce et al.* but which incorporates QE asset purchases more directly and also considers the impact on a wider range of government bonds. They find that UK QE1 reduced government bond rates by 27 bp, although the reduction for 10–15 year bonds is larger, 75 bp. The comparison between these papers highlights two important methodological issues. The first is the importance of controls: [Meaning and Zhu \(2011\)](#) have a smaller set of controls and do not include the fiscal deficit; concern about the deficit in 2009 may have affected UK bond yields in this period. The second is the relative importance of the stock of bonds that is held versus the flow of bond purchases. *Joyce et al. (2011)* consider bond holdings, whereas [Meaning and Zhu \(2011\)](#) consider the impact of bond purchases. The former view implies persistent effects of QE purchases, whereas the latter suggests that the effects of QE are temporary and diminish as QE purchases cease (the implications are discussed further in [Davies \(2011\)](#)). The stark contrast between the 100 bp reduction in *Joyce et al. (2011)* and the more modest 27 bp reduction found by [Meaning and Zhu \(2011\)](#) led to much commentary (e.g. ‘Bank of England said to have over-estimated QE boost’, *Financial Times*, 11 December 2011), highlighting the political sensitivity around QE.

Early analysis suggests that the US QE2 programme may have had a smaller impact on bond yields, albeit a larger effect than found in event studies. [Hamilton and Wu \(2011\)](#) estimate a sophisticated model of bond pricing that incorporates a preferred habitat assumption into an affine arbitrage-free model to motivate an effect of the supply of bonds on prices; they find a reduction of 13 bp on average. [Meaning and Zhu \(2011\)](#) estimate a reduction of 21 bp across the yield curve (but a surprisingly large reduction of 108 bp for 20-year bonds). [Szczerbowicz \(2011\)](#) estimates a small and insignificant effect from US QE2, although the sample here is small.

An interesting strand of work considers how QE in one country affected bond rates in other countries. [Neely \(2011\)](#) analyses the impact of US QE1 on foreign 10-year government bond rates using daily data. He finds a considerable impact on foreign bonds rates, with falls of 65, 78, 54, 50, and 19 bp in the UK, Australia, Canada, Germany, and Japan, respectively. [Glick and Leduc \(2011\)](#) find similar effects, although the effect on UK rates is smaller, at 45 bp. These estimates suggest that the impact of US QE1 on UK bond yields was comparable to that of the UK’s own QE programme and that US QE1 might explain a part of the fall in UK bond rates in late 2008. [Glick and Leduc](#)

(2011) find that UK QE1 had a muted impact on foreign bond rates, with most effects in the low single digits. Given the small scale of this programme in global terms, this finding is not surprising. They also find no effect of US QE2 on foreign bond rates, confirming the limited effect of that programme.

Summarizing the evidence on the impact of QE on government bond rates, event studies find that the initial large-scale asset-purchase programmes (US and UK QE1) did reduce government bond rates, although estimates of the size of the effect vary. Econometric investigations tend to find smaller effects. This may be because there is a strong initial response to announcements of QE programmes which then unwinds over time, an effect that can be detected using econometrics but not event studies. Econometric models estimated on historical data and used to infer the impact of QE estimate larger effects than do models estimated on contemporary data. This may suggest that bond rates are less responsive to policy in periods of economic and financial turmoil, in which case estimates based on historical data should be treated with some caution. The strong impact of US QE1 on overseas bond rates is also notable, suggesting that globally significant QE programmes may have important spill-over effects. The impact of subsequent programmes (US and UK QE2 and Operation Twist) appears to be more modest. There are several possible explanations of this. These programmes may have been expected by the markets and already affected bond rates through anticipation effects. Or there may have been more rapid portfolio adjustment in response to the second round of QE, leading the effects of bond purchases to be transferred more rapidly on to the prices of other assets. A more simple explanation might be that QE programmes are more able to reduce bond rates when those rates are relatively high. Both US and UK QE1 programmes were introduced when the 10-year bond rate exceeded 3 per cent; when subsequent programmes were introduced, this rate was less than 3 per cent. This might be tested in econometric studies by controlling for the bond rate at the start of the programme (cf [Breedon *et al.*, 2012](#)).

III. Evidence of the impact of QE on other asset prices

If the main objective of QE is to affect the real economy, the impact of QE programmes on government bond rates is only part of a chain of causation that connects government bond rates, returns on other assets, aggregate demand, and then output and inflation. In this section we consider evidence on the second stage of this process; the impact of QE on corporate bond rates, money market rates, exchange rates, and equity yields.

As with studies on government bond rates, event studies suggest that QE has a sizeable impact on corporate bond rates. [Krishnamurthy and Vissing-Jorgensen \(2011\)](#) report a strong impact of US QE1 on corporate bond rates with reductions of 77 bp for AAA bonds, 93 bp for A, 60 bp for BA, and 43 bp for B. [Neely \(2011\)](#) reports similar findings: a 78 bp reduction for BAA-rated corporate bonds. [Joyce *et al.* \(2011\)](#) report a smaller effect for UK QE1, a fall of around 50 bp, but only for higher-quality assets at shorter durations. Econometric evidence is more mixed: the estimates of [Joyce *et al.* \(2011\)](#) suggest that UK QE1 led to a reduction of 81 bp in investment-grade corporate rates, falling to 32 bp after 6 months. This effect is similar to the effect estimated for government bond rates. By contrast, the estimates of [Krishnamurthy and Vissing-Jorgensen](#)

(2011) and Wright (2011) for US QE1 and QE2 find a much smaller effect. This may be because Joyce *et al.* use historical data whereas the other papers use contemporary data.

There is little evidence of any impact of QE programmes on money market rates. Neely (2011) finds US QE1 had insignificant effects in the US and a range of other countries. Szczerbowicz (2011) finds no evidence that US QE1 or QE2 had any impact on LIBOR–OIS spreads,¹ in contrast to the clear effect of earlier government bailouts and guarantee schemes. Glick and Leduc (2011) and Neely (2011) consider the impact of US QE1 on exchange rates, concluding that it was associated with depreciations of the US dollar against major currencies in the range of 3–11 per cent. Glick and Leduc (2011) also find that US QE2 had a more negligible effect. They also find a smaller effect associated with UK QE1, with depreciations of sterling in the range of 1.5–3.5 per cent. Joyce *et al.* (2011) find much the same, noting that the slight fall in sterling is much less than the 8 per cent depreciation implied by a simple UIP (uncovered interest rate parity) calculation. The relatively small impacts of US QE1 on exchange rates do not offset the marked impact of the programme on foreign government bond rates; this suggests that a stimulus to economies overseas may be an important but unforeseen aspect of US QE1.

There is less evidence on the impact of QE on other financial assets. Joyce *et al.* (2011) estimate the response of equity yields to UK QE1, falling by 282 bp on impact and by 120 bp after 6 months. Glick and Leduc (2011) find that QE affects commodity prices, with US QE1 reducing commodity prices by 11 per cent, with the effects concentrated in energy-related commodities and precious metals; UK QE1 had a similar but smaller effect. Finally, we note that QE does not appear to have affected interest rates facing small and medium enterprises and households, especially rates on unsecured borrowing. UK evidence (from the database on the Bank of England website) suggests that effective interest rates on this type of borrowing increased in 2008–9 as widening spreads outweighed the effects of lower policy rates.

IV. Evidence of the impact of QE on the real economy

This section considers the impact of QE programmes on output and inflation. Event studies are not feasible since these variables are only observed at monthly or lower frequencies. Evidence comes from econometric studies using data largely drawn from before 2008. We consider six studies (all from within central banks or first published as central bank working papers). Three studies, Baumeister and Benati (2010), Lenza *et al.* (2010), and Kapetanios *et al.* (2012), share a similar methodology, with two others, Peersman (2011) and Bridges and Thomas (2012), sharing elements of the approach. It is therefore worth considering the methodology in some detail.

Consider the following simplified model:

$$Y_t = \Theta_0 + \Theta_1 Y_{t-1} + e_t \quad (1)$$

where $Y_t = (y_t, \pi_t, i_t, s_t)'$ is a (4x1) vector of endogenous variables, comprising measures of output (y), inflation (π), the short-term policy rate (i), and the spread between the policy rate and a longer-term rate (s); e_t is a (4x1) vector of error terms; Θ_0 is a

(4x1) parameter vector; and Θ_1 is a (4x4) parameter matrix, whose elements are Θ_1^{ij} for $i, j \in \{y, \pi, i, s\}$. There are four structural shocks, comprising an aggregate demand shock (ε^d), an aggregate supply shock (ε^s), a conventional monetary policy shock (ε^{cmp}), and unconventional monetary policy shock (ε^{ump}). The reduced form errors are related to the underlying structural shocks by the relationship

$$e_t = B\varepsilon_t \quad (2)$$

where $\varepsilon_t = (\varepsilon_t^d, \varepsilon_t^s, \varepsilon_t^{cmp}, \varepsilon_t^{ump})'$ and B is a (4x4) matrix whose elements are $B^{i,j}$ for $i \in \{y, \pi, i, s\}$ and $j \in \{d, s, cmp, ump\}$. The model is identified by a mixture of exclusion and sign restrictions on B . In particular, the conventional monetary policy shock is defined as moving the policy rate and the spread in opposite directions, while the unconventional monetary policy shock is defined as affecting the spread but not the policy rate.

The effect of QE on output and inflation is then analysed. This is done by comparing forecasts of these variables obtained using this model with forecasts obtained from a counterfactual scenario in which QE did not occur. The counterfactual forecast is constructed by fixing the spread at a value that, it is assumed, would have been observed in the absence of QE. [Baumeister and Benati \(2010\)](#) assume that the spread would have been 60 bp higher in the US (using the mid-point of [Gagnon *et al.*'s, 2011](#), estimated reduction of 38–82 bp) and would have been 50 bp higher in the UK (using [Meier's \(2009\)](#) estimate of a 35–60 bp reduction). [Kapetanios *et al.* \(2012\)](#) assume that the spread would have been 100 bp higher in the UK (using the reduction reported by [Joyce *et al.* \(2011\)](#)). [Lenza *et al.* \(2010\)](#) assume that the spread would have remained fixed at the immediate pre-crisis level.

[Baumeister and Benati \(2010\)](#) use US quarterly data for 1954Q3–2008Q4 and UK quarterly data for 1955Q1–2008Q4 (they also estimate models for the eurozone and Japan). They find that in the absence of QE, output growth would have fallen to -10 per cent compared to an actual fall of 3 per cent and that inflation would have been negative. For the UK, they find that output growth would have fallen to -14 per cent and inflation would have fallen to below -4 per cent. Clearly, these estimates suggest QE had a dramatic impact, saving the UK and US from economic catastrophe. Are they plausible? It is certainly arguable that policy interventions in late 2008 and 2009 prevented a much worse outcome. But QE was only part of a much wider series of interventions that included fiscal policy. These other interventions are not included in the model, so it is possible that the inferred impact of QE captures some of these other effects. The estimates also imply that QE had an immediate impact: for both the US and UK the largest impact of QE is felt in 2009Q1. But UK QE1 did not begin until the middle of 2009Q1 and US QE1 was only expanded from \$600 billion to \$1.75 trillion on 18 March 2009, near the end of the quarter in which QE is estimated to have its largest impact.

[Kapetanios *et al.* \(2012\)](#) use a similar approach to investigate the impact of UK QE1. They use three alternative models. The first is a large-scale VAR estimated using monthly data for April 1993 to September 2011. The model comprises 43 variables, reflecting real activity, prices, and policy rates in the UK, US, and eurozone). This helps

ensure a good forecasting performance, but makes the model vulnerable to the critique that the many relationships built into the model might change as a result of the crisis. The second model is a switching VAR with four regimes. The VAR has 6 variables: output growth, inflation, M4 growth, the 3-month Treasury Bill rate, the spread between the 10-year bond rate and the 3-month Treasury Bill rate, and growth in the FTSE all-share index. Unlike Markov-style models, in this model the economy does not oscillate between regimes; rather it progresses from one regime to another over time. This model is less vulnerable to changes in economic relationships, allowing the use of a longer time period, January 1963 to September 2011. However some regimes are short: for example, the final regime is estimated over the period of the financial crisis. The final model is a VAR in which the parameters are allowed to change over time, evolving as random walks. The VAR has four variables: output growth, inflation, the spread, and the 3-month TB rate, and is estimated using quarterly data over 1968–2011. This model is designed to overcome the problem of economic relationships that evolve over time; however, it is vulnerable to sudden changes in these relationships, which may have occurred during the financial crisis.

They find that UK QE1 increased output growth and inflation; summarizing across models, GDP growth was 150–200 bp higher and inflation 75–150 bp higher. Impulse responses suggest the largest impact on output and inflation, 200 bp and 150 bp, respectively, occurred in March 2010, a month after the end of asset purchases was announced. Given that UK GDP fell by 4.9 per cent in 2009, these results imply that output would have fallen by close to 7 per cent if there had been no QE. Also CPI inflation would have been uncomfortably close to zero rather than its actual value of 2.1 per cent. In short, these results suggest that QE saved the UK from a much deeper recession and the threat of deflation. Although these estimates are lower than found by [Baumeister and Benati \(2010\)](#) (and the timing of the maximal effect more plausible), it might be argued that these estimates are on the high side. The assumption that QE lowered the spread by 100 bp is questionable. As discussed above, most studies find a much smaller effect, close to 50 bp, and most econometric estimates find an effect of less than 50 bp. Perhaps a reduction of 50 bp, following [Baumeister and Benati \(2010\)](#), might have been considered. They also assume that the 100 bp reduction applies across the yield curve: the evidence discussed above suggests there were smaller reductions in shorter-dated bonds.

[Lenza et al. \(2010\)](#) investigate the impact of conventional and unconventional monetary policy initiatives of the ECB using monthly data for January 1991 to August 2009. This is not a study of QE, since the ECB has not opted for large-scale purchases of long-dated financial assets. Rather, it is a study of the effect of large-scale unconventional policy interventions in the short-term money market. They use a large-scale VAR comprising 31 variables covering various aspects of output, unemployment, prices, the money supply, and policy, and other interest rates in the Eurozone. They analyse the impact of unconventional monetary policy on interest rate spreads and use this to infer the impact on the real economy. Their counterfactual scenario implies that unconventional monetary policy had a strong effect on short-term interest rates: for example the 3-month rate is assumed to be close to 3 per cent in mid-2009 compared to an actual rate of less than 1 per cent. The estimated impact on output is initially negative, reducing output growth by up to 100 bp; the effect become positive after mid-2009 and rises to a peak of 260 bp by late 2009 before declining, disappearing after mid-2012. The cumulative effect is positive. The initial impact on inflation is persistently negative, only

becoming positive by March–April 2011, by which time the effect is small. The cumulative effect is negative. The policy simulations suggest that the effect of the policy was transmitted via changes in interest rates spreads rather than through the money supply. The results are mixed: the positive impact on output is offset by the finding that inflation was reduced; this is an odd finding for a policy that is assumed to have reduced interest rates by up to 200 bp. The finding of little change in the broad money supply is also a little odd: given the importance of bank lending in the Eurozone, one might have expected a positive impact on output to be reflected in a higher money supply.

These studies all measure the impact of QE by assuming a counterfactual value for the interest rate spread that would have been observed in the absence of QE. The counterfactuals that are used are open to criticism: in particular, they assume that QE had a strong and lasting impact on government bond rates and hence on the spread. The evidence surveyed above suggests that initial QE programmes had a strong initial impact on bond rates but that subsequent programmes had little effect. Econometric evidence also suggests that even in the early programmes, the strong initial impact was only short lived. By assuming a strong impact of QE on the spread throughout the period, these studies may be assuming QE had a stronger effect than it actually did (although it might be argued that [Joyce *et al.* \(2011\)](#) under-estimate the impact of QE, since they do not allow for anticipation effects).

[Bridges and Thomas \(2012\)](#) analyse the impact of UK QE1 from a more explicitly monetarist perspective. They argue that QE increased UK M4 by around £120 billion or 8 per cent. They analyse the impact of such an increase using an eight-variable VAR, estimated using UK quarterly data for 1964Q1–2007Q3 and comprising the level of output, inflation, the real exchange, the money supply, stock prices, a short-term interest rate, and the rate on government bonds. There are eight shocks including conventional and unconventional monetary policy shocks, distinguished by the assumption that unconventional policy does not affect the short-term rate. The effects of QE are simulated by choosing unconventional monetary policy shocks that result in a permanent 8 per cent increase in the nominal money supply beginning in 2009Q1, setting the other shocks to zero. Simulations suggest that QE reduced the bond rate spread by a maximum of 175 bp by the start of 2010, a much larger effect than in other studies. However, this effect unwinds, becoming zero by mid-2011 and stabilizing at a 50 bp increase. Equity prices increase by 200 bp by early 2010; this effect then also unwinds and stabilizes at a 50 bp increase. The nominal exchange rate depreciates, but this effect is almost exactly offset by higher prices. The response of output is more modest. The level of output is projected to begin increasing in mid-2009, rising to a maximum effect of around 150 bp by the start of 2010. The effect then declines and becomes negligible by mid-2011. Inflation increases, but only from early 2010. It reaches a peak of around 200 bp before declining, although remaining above zero throughout the forecast period.

As the authors acknowledge, these results assume that a model that is estimated using pre-crisis data and that is simulated using much larger shocks than occurred in the estimation period can deliver reliable insights into the impact of QE. The very strong response of asset prices, especially bond rates, is perhaps implausible in the light of other studies. That said, the estimated response of output and inflation to QE seem more plausible than some of the larger effects obtained in some other studies.

[Peersman \(2011\)](#) considers the impact of policy interventions on the monetary base and money multiplier, and investigates transmission to the real economy via this route

rather than via interest rate spreads. In contrast to the papers mentioned above, he assesses the impact of unconventional monetary policy by analysis of the impact of identified unconventional policy shocks rather than by comparison with a counterfactual no-intervention case. Using monthly data for the eurozone for 1991M1–2009M12, he analyses a VAR that includes industrial production as a proxy for output, the (CPI) price level, total credit, the monetary base, the interest rate on credit (an average of various private-sector lending rates), and the policy rate. Conventional policy shocks have hump-shaped effects on output and inflation. Unconventional monetary policy shocks have similar effects, although these occur several months later. This similarity leads [Peersman \(2011\)](#) to suggest that the two forms of monetary policy can to some extent be seen as substitutes in terms of their impact on the real economy, with an unconventional monetary policy shock that increases the monetary base by 10 per cent having the same impact as a 25 bp cut in the policy rate. However, there are differences in the transmission mechanism, with conventional policy widening credit spreads and increasing the money multiplier, and unconventional policy having the opposite effect.

[Chung *et al.* \(2011\)](#) use a different approach. They estimate the reduction in government bond rates due to increased Federal Reserve holdings of government bonds as part of QE and simulate the impact of this reduction in bond rates using the large-scale Federal Reserve Board/US macroeconomic model. They assume that the impact of QE unwinds over time, from an initial reduction of 50 bp that reduces to a 30 bp reduction by late 2010 and a 10 bp reduction by late 2012. They find that US QE1 boosted GDP growth by almost 200 bp, with the strongest effect felt in early 2010, and that US QE2 increased GDP growth by around 100 bp. The effect is similar to that of a 300 bp reduction of the policy rate.

Econometric estimates of the effects of QE programmes on output and inflation (with the exception of [Baumeister and Benati, 2011](#)), are broadly similar. QE increased GDP growth by around 1–3 per cent with a similar effect on inflation. This evidence suggests that QE has proved effective in limiting the scale of the downturn caused by the financial crisis that began in 2007. However, it suggests that QE, by itself, is not strong enough to spark an economic recovery. Of course, this conclusion is tentative. There is relatively little evidence on the effects of QE and most of this uses the same methodology.

V. Conclusions

What have we learnt? We have stressed the difficulty of assessing the impact of QE and the compromises forced on investigations of these issues. Given this, we doubt it is possible to achieve any consensus. However, some consistent conclusions do emerge from our survey of the evidence.

First, event studies suggest that initial large-scale QE programmes did succeed in lowering government bond rates, especially at the longer end of the yield curve. However, econometric studies suggest that these effects may have been only temporary. By contrast there is little evidence of subsequent QE programmes having much effect. This may be because they were introduced when bond rates were already quite low. Or it may be because initial QE programmes were sufficient to demonstrate the commitment

of policy-makers to addressing the financial crisis, leaving little for subsequent programmes to add. Second, QE appears to have been an effective response to the severe economic difficulties of late 2008 and 2009, preventing even larger declines in output and inflation than were experienced. However, QE is a rather weak policy instrument; the very large initial asset-purchase programmes had effects comparable to a reduction of 200–300 bp in the policy rate; subsequent programmes had little effect. Third, (with one exception) studies of the impact of QE on the real economy use the same methodology. Alternative estimates based on different models and methodologies would be useful in widening an evidence base that is currently too narrow.

As we approach the fifth anniversary of the first US QE programmes in 2008, issues of how policy-makers might unwind QE are likely to become more prominent. Large-scale sales of government bonds might threaten the stability of an already fragile market and lead to rapid increases in bond rates. This suggests that reduction in the size of central bank balance sheets is likely to be done more gradually. This may create difficulties for policy-makers, who may need to moderate or delay increases in the policy rate to absorb the effect of bond sales. Given continuing uncertainties in the eurozone and slow progress in reducing fiscal deficits, it may well be some years before QE begins to be reversed.

References

- Baumeister, C., and Benati, L. (2010), 'Unconventional Monetary Policy and the Great Recession—Estimating the Impact of a Compression in the Yield Spread at the Zero Lower Bound', ECB Working Paper No. 1258.
- Bridges, J., and Thomas, R. (2012), 'The Impact of QE on the UK Economy—Some Supportive Monetarist Arithmetic', Bank of England Working Paper, No 442.
- Breedon, F., Chadha, J. S., and Waters, A. (2012), 'The Financial Market Impact of UK Quantitative Easing', *Oxford Review of Economic Policy*, **28**(4), XXX.
- Chung, H., Laforge, J. P., Reifschneider, D., and Williams, J. (2011), 'Have We Underestimated the Likelihood and Severity of Zero Lower Bound Events?', Federal Reserve Bank of San Francisco Working Paper No. 2011-01.
- D'Amico, S., and King, T. (2010), 'Flow and Stock Effects of Large-scale Treasury Purchases', Board of Governors of the Federal Reserve System Finance and Economics Discussion Series No. 2010-52.
- Davies, G. (2011), 'Can Bonds Survive the Fed's Exit?', *The Financial Times*, 30 March.
- Gagnon, J., Raskin, M., Remache, J., and Sack, B. (2011), 'The Financial Market Effects of the Federal Reserve's Large-scale Asset Purchases', *International Journal of Central Banking*, **7**, 3–43.
- Glick, R., and Leduc, S. (2011), 'Central Bank Announcements of Asset Purchases and the Impact on Global Financial and Commodity Markets', Federal Reserve Bank of San Francisco Working Paper No. 2011-30.
- Hamilton, J. D., and Wu, J. C. (2011), 'The Effectiveness of Alternative Monetary Policy Tools in a Zero Lower Bound Environment', NBER Working Paper No. 16956.
- Joyce, M., Lasasosa, A., Stevens, I., and Tong, M. (2011), 'The Financial Market Impact of Quantitative Easing in the United Kingdom', *International Journal of Central Banking*, **7**, 113–61.
- Kapetanios, G., Mumtaz, H., Stevens, I., and Theodoridis, K. (2012), 'Assessing the Economy-wide Effects of Quantitative Easing', Bank of England Working Paper No 443.
- Krishnamurthy, A., and Vissing-Jorgensen, A. (2011), 'The Effects of Quantitative Easing on Long-term Interest Rates', Northwestern University, mimeo.

- Lenza, M., Pill, H., and Reichlin, L. (2010), 'Monetary Policy in Exceptional Times', *Economic Policy*, **25**, 295–339.
- Meier, A. (2009), 'Panacea, Curse, or Nonevent: Unconventional Monetary Policy in the United Kingdom', IMF Working Paper No. 09/163.
- Meaning, J., and Zhu, F. (2011), 'The Impact of Recent Central Bank Asset Purchase Programmes', *Bank of International Settlements Quarterly Review*, December, 73–83.
- Neely, C. (2011), 'The Large-scale Asset Purchases Had Large International Effects', Federal Reserve Bank of St Louis Working Paper No. 2010-018C.
- Peersman, G. (2011), 'Macroeconomic Effects of Unconventional Monetary Policy in the Euro Area', CESifo Working Paper No. 3589.
- Szczerbowicz, U. (2011), 'Are Unconventional Monetary Policies Effective?', Working Paper 1107, LUISS Guido Carli.
- Swanson, E. (2011), 'Let's Twist Again: A High-frequency Event-study Analysis of Operation Twist and its Implications for QE2', *Brookings Papers on Economic Activity*, Spring.
- Wright, J. (2011), 'What Does Monetary Policy Do to Long-term Interest Rates at the Zero Lower Bound?', NBER Working Paper 17154.